GREEN CODE HACKATHON 2026 | WATER AND CLIMATE RESILIENCE

Mvula

One WhatsApp early-warning and climate-intelligence platform for water, floods and farms in Zambia.

Merged proposal combining the Mvula, Rainward and AgriPal concepts | 31 October 2026

1. Executive summary

Three separate ideas turned out to be the same product seen from different angles. **Mvula** warns Lusaka neighbourhoods about water shortages and floods. **Rainward** turns climate signals and farmer observations into decisions for agricultural advisers. **AgriPal** turns a farm's coordinates, weather and soil data into a crop recommendation. All three take public climate and utility data, tie it to a specific place, reach people through a familiar channel, and need ground-truth feedback to stay accurate.

We propose a single platform, keeping the name **Mvula** (Nyanja for "rain"). It has one WhatsApp channel, one location-based risk engine and one verified-report pipeline, with two modules on top: **Mvula Water and Flood** for urban compounds and **Mvula Farm** for crop and dry-spell advice. The hackathon build focuses on the Water and Flood module, because the rains start now and it makes the strongest live demo. Mvula Farm reuses the same engine and the existing crop model, so it adds little extra build time.

2. The problem

- **Water:** Lusaka needs about 480,000 cubic metres a day and produces about 250,000. Roughly 47% of supply comes from boreholes, which stop when load shedding cuts power. Families do not know when to fill containers.
- **Floods:** low-lying compounds such as Kanyama flood most seasons, and floods bring cholera. The 2023 floods displaced nearly 400,000 people. Warnings are national and generic, not "your street, tonight".
- **Farms:** choosing a crop and responding to a dry spell depends on soil, rainfall, temperature and location, and many farmers cannot get that analysis quickly. Advisers cannot visit every farm every day.
- **Common root:** the data exists but is scattered, is not localised, and there is no live feedback from the ground. Institutions (Lusaka Water, the council, DMMU, NGOs, consultants) lack a street- or farm-level picture.

3. What we kept from each concept

To keep the merger focused, we took only the fundamental idea from each concept and dropped anything that is not needed to prove the core loop.

Concept	Fundamentals kept	Deliberately left out (for now)
Mvula	WhatsApp as the only resident interface (no app to install). Ward-level risk score refreshed hourly. One-tap and photo community reports. Parser for ZESCO and Lusaka Water schedules. Live ward map. Free for residents, paid for institutions.	Full Nyanja and Bemba coverage and business referral fees (phase 2 or later).
Rainward	The loop: detect, investigate, collect, localise, recommend, learn. Professional-first customer model. Farmers and residents as field-data contributors. Provenance on every observation (who, when, where, reviewed or not). Human review of high-impact advice. Measured data kept separate from reported data.	Research-retrieval (RAG) workspace, Google Earth Engine at launch, multi-role workspace with report builder.
AgriPal	Coordinates in, environmental data out: automatic weather and soil lookup for a location. Random Forest crop recommender (22 crops). Django and REST backend, accounts with several saved locations, admin dashboard, secrets kept in environment files.	Standalone Android app, Gmail SMTP contact form, Kaggle-only training (to be validated against Zambian data).

4. The merged solution

Mvula continuously combines forecasts, satellite rainfall, power and water schedules, soil data and community reports into a risk score for each place (a ward, compound or farm). It sends plain-language alerts on WhatsApp and shows a live map to institutions. Every report from a resident or farmer improves the picture for everyone.

1 DETECT	2 COLLECT	3 LOCALISE	4 ACT	5 LEARN
Forecasts, satellite rainfall, power and water schedules	WhatsApp reports and photos, with timestamp and location	Risk per ward or farm, hourly	Alerts to residents; verified hotspots to institutions	Follow-up check-ins and outcomes refine the model

Two modules on one engine

	Mvula Water and Flood (hackathon focus)	Mvula Farm (reuses the engine)
Users	Residents of Kanyama, Mtendere, Chawama; Lusaka Water, council, DMMU, NGOs	Farmers and farm workers; agricultural consultants and agribusinesses
Alerts	When to fill containers, power and rationing windows, flood and cholera risk tonight, what to do	Crop suggestion for the farm's coordinates; dry-spell and heat warnings; practical guidance
Reports	Dry tap, burst pipe, blocked drain, flooding	Soil moisture, crop condition, water availability, photo
Review	Cross-checked against other reports and weather data before it moves the map	Consultant approves or edits advice before it is sent

Example alert (WhatsApp)

> **Mvula (Kanyama):** Heavy rain expected tonight, about 45 mm. Flood risk HIGH near the main drain. Move valuables up, keep children away from water, boil drinking water. Reply 2 to report flooding.

5. Technical approach

We chose the stack that the team already knows (Python, Django, PostgreSQL) and that the strongest parts of all three concepts already point to, so the hackathon time goes into the product and not into setup.

Layer	Choice	Why
Channel	WhatsApp Business Cloud API; English first, Nyanja and Bemba next	No install, familiar to users, supports text, photos and voice notes
Backend	Python with Django and Django REST Framework; Celery and Redis for hourly jobs	Reuses the AgriPal codebase (accounts, admin, REST endpoint)
Database	PostgreSQL with PostGIS	Ward and farm geometry, spatial joins, report history
Data	Open-Meteo and OpenWeatherMap (forecasts); satellite rainfall; iSDAsoil (soil); schedule parser for ZESCO and Lusaka Water notices	Weather and soil lookup is already built in AgriPal; iSDAsoil replaced SoilGrids, which returned nothing for Zambian cities
Risk and ML	Transparent rule-based risk scores for flood and water; Random Forest for crop recommendation	Deterministic, explainable numbers; ML only where a trained model exists
AI use	LLM used only to phrase alerts in plain language and in local languages, grounded in computed values	An LLM must never invent rainfall figures or certainty
Dashboard	Leaflet map with OpenStreetMap tiles; institution and consultant views with role-based access	Already proven in AgriPal; live map is the demo centrepiece
Hosting	Google Cloud, secrets in environment variables or Secret Manager, code in GitHub with tagged releases	Matches hackathon track; avoids the data-loss and leaked-credential risks AgriPal recorded

Trust rules built in from day one

- Every report stores who sent it, when, where, by what method, and whether it has been reviewed. Reports are never presented as measurements.
- Alerts show the source type: forecast, schedule, or community report, and say when confidence is low.
- Alerts about health or evacuation, and all farm advice, go through a human reviewer until the model is validated.
- Residents and farmers give consent for location, photos and voice notes, and can correct or delete their data. Public maps show aggregates only.

6. Users and revenue model (to be validated)

Following Rainward, we are professional-first: those who need the aggregated picture pay, and the people who supply ground data pay nothing.

Segment	Value	Offer
Residents and farmers	Timely alerts, advice, a voice in the data	Free, always
Lusaka Water, council, DMMU	Live faults, hotspots and flood risk by ward	Dashboard subscription
NGOs and health programmes	Targeted water, sanitation and cholera response; community climate monitoring	Programme licence (donor-funded)
Consultants, agribusinesses, researchers	Monitor many farms, prioritise visits, structured field data	Subscription by users, farms or area
Schools, clinics, lodges, water kiosks	Plan water and protect stock	Paid site alerts; referral fees from tank and bowser delivery

Over time the platform builds a dataset linking climate signals to ground conditions, actions and outcomes. That history is the long-term advantage, and it improves with each season.

7. Delivery plan

48-hour hackathon scope

- Live WhatsApp bot: subscribe, choose area, receive a real alert (flood and water).
- Ward map of Lusaka with risk driven by real forecast data.
- Current power and water schedules loaded for at least three compounds.
- Live stage demo: an audience member sends a flood report and the map updates.
- Stretch: send a location pin to the bot and get a crop suggestion from the AgriPal model, to show the second module.
- Interviews with 15 or more residents and feedback from water and disaster experts.

After the hackathon

Phase	Timing	Outcome
Pilot: water and flood	First 90 days, rainy season	Three compounds, partner with one utility or DMMU, 500+ subscribers, alert accuracy tracked against reports
Pilot: farm	Same period, in parallel at small scale	One district, one consultant, 20 to 50 farmers; drought check-in cycle works end to end; crop model validated against local advice
Harden	Months 4 to 6	Full PostgreSQL and PostGIS data model, secure email and messaging, tests, review workflow, Nyanja and Bemba
Scale	After first season	More wards and districts, institutional contracts, optional lightweight mobile client if data shows a need

8. Team and roles

Role	Responsibility
Backend developer	WhatsApp bot, data pipelines, schedule parser, Django API (AgriPal code reused)
Data scientist or GIS analyst	Ward and farm risk model, crop model validation, map layers
Frontend developer	Live dashboard and map, institution and consultant views
Water, health or climate specialist	Alert wording, actions, thresholds, review rules
Business lead and designer	Interviews, partnerships, pricing tests, pitch

9. Success measures and risks

Measure	Target to be confirmed in pilot
Subscribers and active alert recipients	Growth and retention per compound or district
Report volume and share of reports that are complete and usable	Majority of reports usable; low time to map update
Alert quality	Flood and shortage flags matched by later verified reports
Institutional uptake	At least one utility, council or NGO using the dashboard
Farm module	Check-in completion rate; farmer understanding and trust; consultant time saved

Risk	Mitigation
Utility schedules are published as notices in varying formats and may change	Start with three compounds, parse manually where needed, allow residents to correct schedules
False or malicious reports	Provenance, cross-checks, human review, rate limits
A crop model trained on a generic dataset may not fit Zambian conditions	Label as suggestion, validate with consultants, retrain on local data
Soil data access (iSDAsoil credentials pending)	Fall back to regional soil defaults and say so in the output
Scope creep across two modules	Water and Flood first; Farm limited to one pilot workflow
Lost work or leaked keys	GitHub as source of truth with tagged milestones; secrets never in version control
Privacy of location and photos	Consent, role-based access, aggregated public views

10. Next steps

- Agree the name, repository and branch structure; start from the AgriPal Django codebase and tag v1.
- Confirm WhatsApp Business API access and iSDAsoil credentials today.
- Assign the five roles and split the 48 hours: data and bot first, map second, demo last.
- Book resident interviews in at least two compounds before the demo.

Sources used in the original concepts: Zambia Monitor (Lusaka water demand and production, borehole pumping and power); Lusaka Times (LWSC rationing, 53% Kafue River and 47% boreholes); Crisis24, March 2023 (Lusaka floods, about 400,000 displaced). Pricing and data-source availability are assumptions to be tested during the hackathon and pilot.